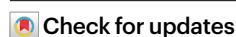


Ambient-pressure conversion of plastic waste to jet fuel cycloalkanes by tandem hydropyrolysis and vapour-phase hydrogenation

Received: 30 August 2023

Accepted: 27 April 2026

Published online: 27 May 2026



Jia Wang^{1,13}, Zedong Zhang^{2,13}✉, Shule Wang^{3,4,13}, Ya-Fei Jiang^{5,6,13}, He Zhou^{7,13}, Wenhui Zhong^{8,13}, Yiyun Zhang^{1,9}, Dongxian Li¹, Wenjun Zhong¹⁰, Shiyao Li¹¹, Daniel L. Sanchez¹²✉, Dingsheng Wang¹²✉, Jun Li^{5,6,12}✉, Jianchun Jiang^{1,9}✉ & Yadong Li²✉

Converting plastic waste into jet fuel could support the decarbonization of the aviation industry, yet current upcycling routes rely on high pressures (~3 MPa) and prolonged reaction times (up to 144 h). Here we report a tandem hydropyrolysis and vapour-phase hydrogenation strategy enabled by a single-atom Ru catalyst on Co-Al oxides ($\text{Ru}_{\text{SA}}/\text{CoAlO}_x$). The catalyst achieves a turnover frequency of 144 s^{-1} for benzene hydrogenation at atmospheric pressure, exceeding that of commercial Ru/C by over 100-fold. In a tandem fixed-bed reactor with hydropyrolysis at 460 °C and downstream vapour-phase hydrogenation at 160 °C, polystyrene is converted to 94.8 wt% cycloalkanes at 0.15 MPa and to 59 wt% cycloalkanes at atmospheric pressure. The approach extends to mixed plastics, achieving jet-fuel-range hydrocarbon yields above 82 wt%. The catalyst remains stable for over 110 h during continuous vapour-phase hydrogenation. Life-cycle analysis indicates a 73% reduction in well-to-pump CO_2 emissions relative to petroleum-based jet fuel, and techno-economic analysis suggests a competitive minimum selling price of US\$1.0–1.8 per kg.

Cycloalkanes are indispensable components of transportation fuels, particularly in aviation, where their high heat-sink capacity is critical for managing the extreme thermal loads of high-Mach-number flight^{1–4}. Beyond fuels, cycloalkanes also serve as key building blocks in chemicals and materials^{5–8}. However, conventional production of cycloalkanes from petroleum-derived aromatics typically requires energy-intensive conditions (>200 °C, ~2 MPa), posing a major barrier to low-carbon fuel production⁹. This challenge is further compounded by the low global plastic recycling rate (<9%), underscoring the need for strategies that can convert waste plastics into value-added fuel-range

products^{10–12}. Chemical upcycling, including catalytic hydrogenolysis and pyrolysis-based routes, therefore offers a promising pathway towards aviation fuel production^{13–16}.

Hydrogenolysis, which integrates polymer chain cleavage and hydrogenation in a single step, has been widely explored for plastic upcycling¹⁷. However, hydrogenolysis typically requires long reaction times (up to 144 h) and elevated hydrogen pressures (~3 MPa), resulting in low process efficiency and high operational cost^{18–22}. Although Ru-based catalysts often show high activity, they generally require high metal loadings, which further increase catalyst cost^{23–26}. In addition,

A full list of affiliations appears at the end of the paper. ✉e-mail: zedong.zhang@northwestern.edu; wangdingsheng@mail.tsinghua.edu.cn; junli@tsinghua.edu.cn; jiangjc@icfp.cn; ydli@mail.tsinghua.edu.cn

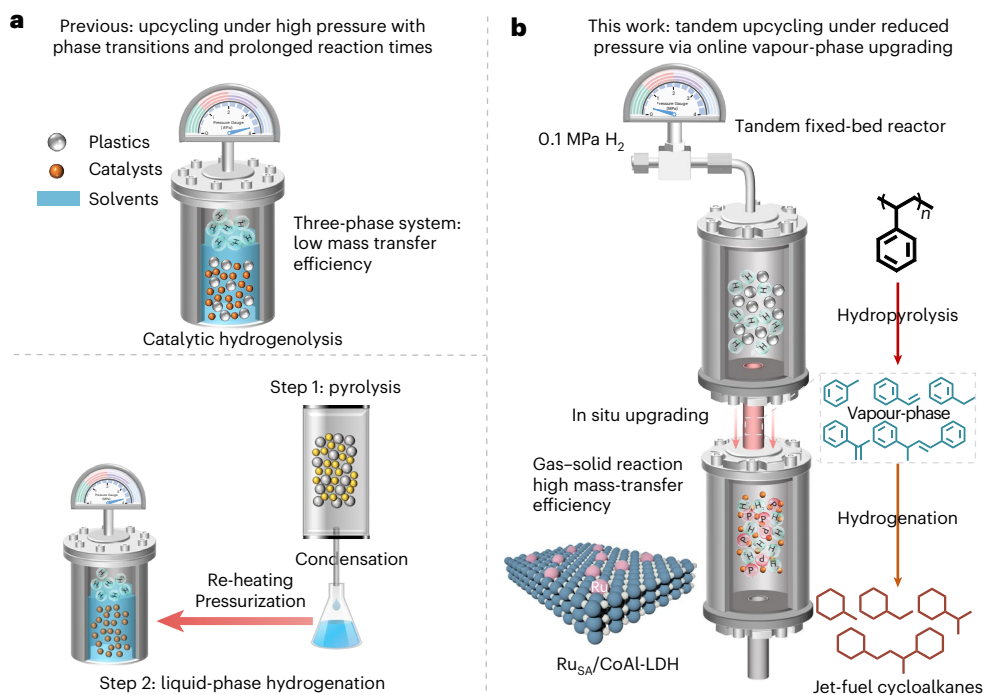


Fig. 1 | Process design for plastic upcycling to jet-fuel-range cycloalkanes.

a, Schematic comparison of conventional plastic upgrading routes, including direct catalytic hydrogenolysis in a three-phase system and pyrolysis followed by condensation, reheating, repressurization and liquid-phase hydrogenation.

b, Proposed tandem upcycling strategy integrating hydropyrolysis with downstream vapour-phase hydrogenation in a fixed-bed reactor, enabling in situ upgrading in a gas–solid regime without intermediate condensation and repressurization. Figure adapted with permission from ref. 48, Springer Nature Limited.

hydrogenolysis has been mainly demonstrated for single plastic types, whereas real-world plastic waste is compositionally heterogeneous^{27–35}. By contrast, pyrolysis is an efficient and industrially relevant route for processing mixed plastic streams because it separates polymer depolymerization from subsequent upgrading³⁶. However, pyrolysis products contain unsaturated and oxygenated species that are prone to secondary reactions and therefore require downstream upgrading, most commonly via liquid-phase hydrogenation^{37,38}. This additional step increases process complexity and is limited by multiphase mass transfer (Fig. 1a), hindering the efficient production of saturated fuel-range hydrocarbons^{39,40}. Integrating pyrolysis with vapour-phase hydrogenation in a tandem configuration offers a route to overcome these limitations by separating depolymerization from upgrading while avoiding intermediate phase transitions (Fig. 1b). Such configurations simplify process design and improve overall process efficiency, but their practical implementation depends on catalysts capable of delivering efficient and selective hydrogenation under mild conditions.

Here we report a tandem strategy that combines hydropyrolysis with vapour-phase hydrogenation and enables direct upgrading of plastic-derived intermediates over Ru single atoms supported on $CoAlO_x$. The process affords jet-fuel-range cycloalkanes in up to 94.8 wt% yield at 0.15 MPa and still delivers 59.4 wt% cycloalkanes at atmospheric pressure. It is applicable to mixed plastic feedstocks and maintains stable performance under continuous operation. Life-cycle and techno-economic analyses indicate reduced CO_2 emissions and competitive production costs, supporting the scalability of this strategy for producing jet-fuel-range cycloalkanes from plastic waste.

Catalyst structure characterization

The successful synthesis of our designed catalyst was confirmed by the presence of well-dispersed Ru single atoms on a lamellar metal oxide. We used energy-dispersive spectrometry (EDS) mapping to evaluate the distribution of Ru atoms. Ru atoms are atomically dispersed on $CoAlO_x$, as indicated by the yellow dots in Fig. 2a. Aberration-corrected

high-angle annular dark-field scanning transmission electron microscopy further revealed Ru atoms within the crystal defects of $CoAlO_x$ (Fig. 2b). A magnified image in Fig. 2c offers a closer view of these crystal defects, where blue circles indicate defects in the support and highly contrasted yellow circles denote Ru single atoms. This characterization confirms the atomic-level dispersion of Ru atoms within the $CoAlO_x$ support.

We collected powder X-ray diffraction spectra to confirm the composition of the $CoAlO_x$ support. The analysis revealed that the support primarily consists of CoO (Supplementary Fig. 1). Additionally, through a combination of EDS and X-ray photoelectron spectroscopy, we determined that aluminium is distributed within the CoO matrix rather than forming Al_2O_3 (Supplementary Fig. 2). To gain a deeper understanding of the distribution and microscopic environment of Ru in the catalyst, we analysed the $CoAlO_x$ support and the coordination structure of Ru using X-ray absorption fine structure spectroscopy (Supplementary Fig. 3). The X-ray photoelectron spectra indicated that the predominant valence state of Co in the $CoAlO_x$ support was +2, as shown by the presence of CoO (Supplementary Fig. 4). However, upon incorporation of Ru single atoms, the Co $2p_{3/2}$ peak in $Ru_{SA}@CoAlO_x$ shifted by 0.2 eV, indicating a change in the electronic environment of Co. The valence state of Al was determined to be +3, whereas the oxygen peaks were divided into lattice oxygen and oxygen vacancy peaks. For an accurate determination of the valence state of Ru in $Ru_{SA}@CoAlO_x$, we used a 19-element solid-state detector to identify the Ru single atoms and performed X-ray absorption near-edge structure analysis on a reference Ru material (Fig. 2d). Our findings indicate that the oxidation state of the Ru single atoms in $Ru_{SA}@CoAlO_x$ falls within the range of +2 to +3, suggesting partial oxidation and a mixed valence state within the catalyst.

Through Fourier transform analysis of k-space curves, we observed a lack of periodicity in the high k region for the $Ru_{SA}@CoAlO_x$ catalyst (Supplementary Fig. 5), suggesting disorder in the outer shell layer. In the R-space spectra, peaks in the range of 1–2 Å for the $Ru_{SA}@CoAlO_x$

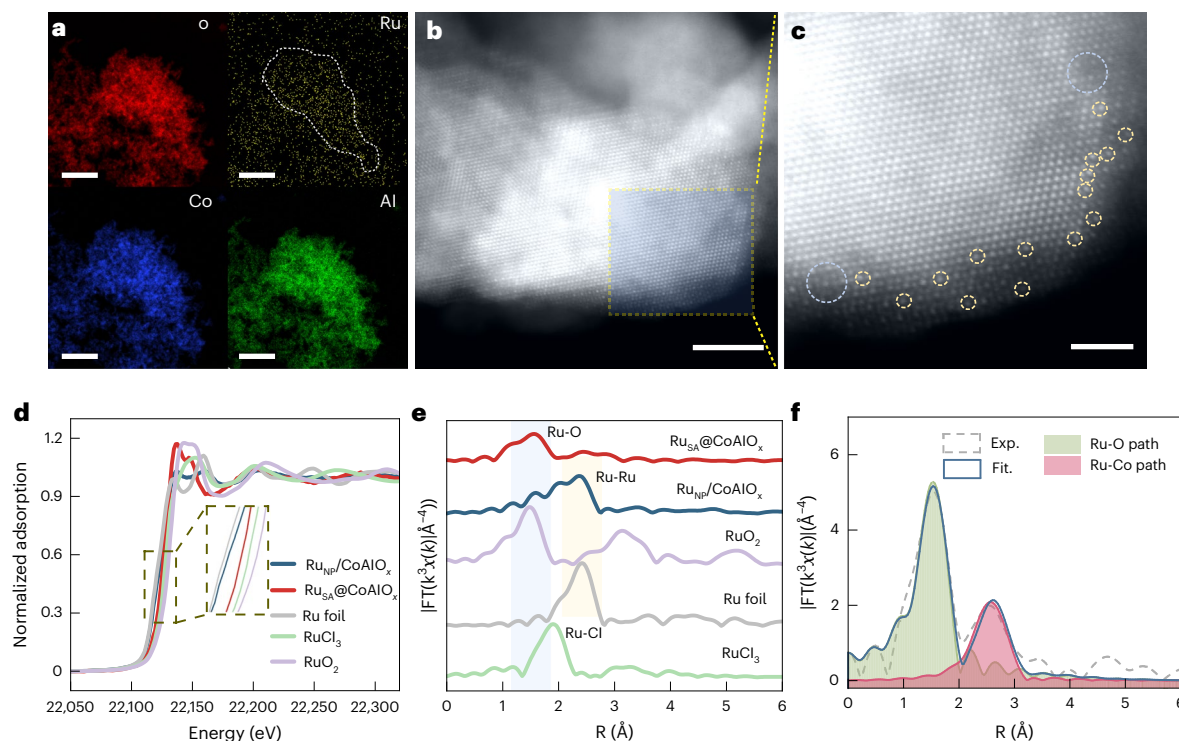


Fig. 2 | Structure and morphology characterization. **a**, Energy-dispersive X-ray spectroscopy mapping showing oxygen (red), aluminium (green), ruthenium (yellow) and cobalt (blue) in $\text{Ru}_{\text{SA}}@\text{CoAlO}_x$. Scale bar: 200 nm. **b**, Annular dark-field scanning transmission electron microscopy image of $\text{Ru}_{\text{SA}}@\text{CoAlO}_x$. Scale bar: 5 nm. **c**, Magnified STEM image showing atomic-scale details. Scale bar: 2 nm. Yellow circles indicate Ru single atoms with higher intensity, and blue circles

indicate defects on CoAlO_x . **d**, Normalized X-ray absorption near-edge structure spectra of $\text{Ru}_{\text{SA}}@\text{CoAlO}_x$ and reference Ru catalysts. **e**, Radial distribution function (R-space) curves of Ru references, highlighting the first-shell Ru–O path (blue box) and the second-shell Ru–Ru path (yellow box). **f**, Fitting results for $\text{Ru}_{\text{SA}}@\text{CoAlO}_x$, the fitting range is 3–12.7 $\text{\AA}^{-1}$, and R-range is 1–3.4 $\text{\AA}$.

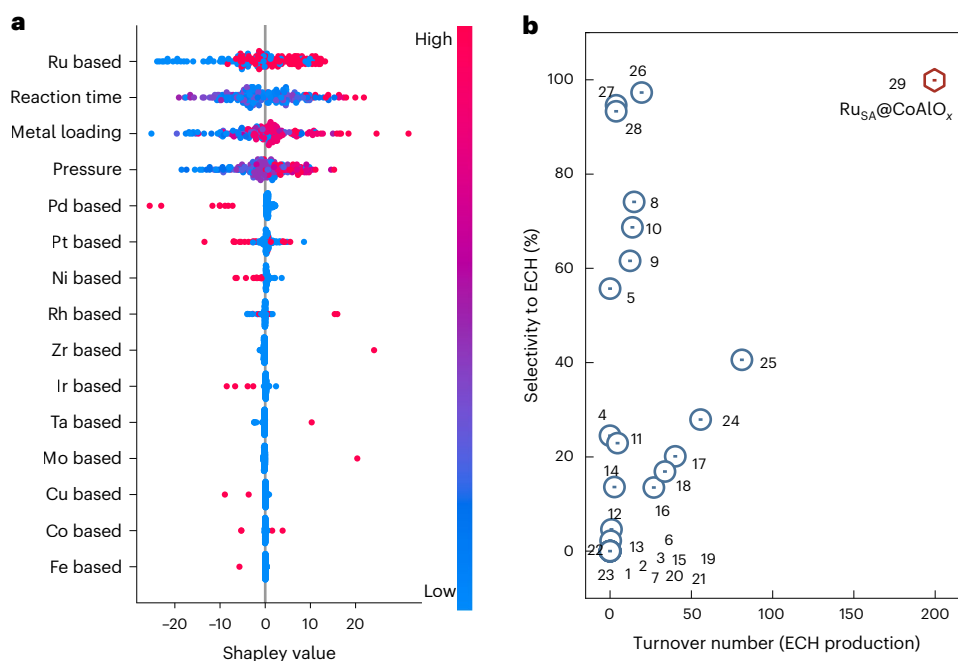


Fig. 3 | Machine learning-guided analysis of key factors and catalytic performance for selective hydrogenation towards jet-fuel-range cycloalkanes. **a**, SHAP Additive exPlanations (SHAP) summary plot showing the contributions of reaction time, pressure, active metal type and metal loading to the model-predicted liquid fuel yield during plastic waste hydrogenolysis. Each point represents an individual data sample from the literature-derived dataset. The x axis (Shapley value) indicates the contribution of each feature to the model prediction relative to the average prediction, with positive values

increasing and negative values decreasing the predicted yield. The colour scale represents the original feature value, ranging from low (blue) to high (pink). For one-hot-encoded metal-type variables, high and low correspond to the presence and absence of the respective metal, respectively. **b**, Selectivity to ethylcyclohexane (ECH) (%) plotted against turnover number (TON) for ECH production during vapour-phase hydrogenation. Each numbered point corresponds to an individual experimental entry from this work, as listed in Supplementary Table 3.

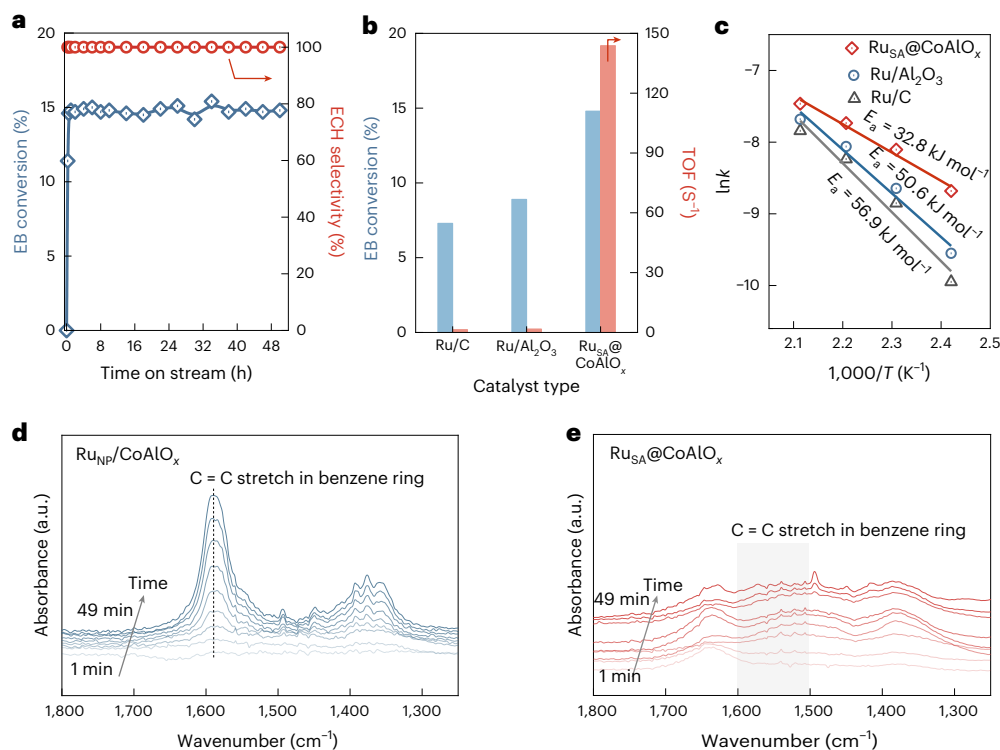


Fig. 4 | Continuous-flow vapour-phase hydrogenation of ethylbenzene and mechanistic investigation. **a**, Time-on-stream profiles of ethylbenzene (EB) conversion and ECH selectivity during continuous vapour-phase hydrogenation over $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$. **b**, Comparison of EB conversion and turnover frequency (TOF) for ECH formation over Ru/C , $\text{Ru}/\text{Al}_2\text{O}_3$ and $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$. **c**, Arrhenius plots for ethylbenzene hydrogenation over Ru/C , $\text{Ru}/\text{Al}_2\text{O}_3$ and $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$. **d,e**, Time-resolved operando diffuse reflectance infrared Fourier transform

spectroscopy (DRIFTS) spectra during ethylbenzene hydrogenation over $\text{Ru}_{\text{NP}}/\text{CoAlO}_x$ (**d**) and $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$ (**e**), showing the evolution of the C=C stretching band of the benzene ring. Unless otherwise specified, reaction conditions for **a–c** were 160 °C and 0.1 MPa H_2 , using 1.8 g of catalyst with a bed volume of 2 ml; the liquid hourly space velocity (LHSV) of ethylbenzene was 1.8 h^{-1} , and the gas hourly space velocity (GHSV) of ethylbenzene vapour was 530 h^{-1} (calculated using the ideal gas law) and the GHSV of H_2 was 1,500 h^{-1} .

indicate bonds within the first shell layer (Fig. 2e). Additionally, peaks around 2.7 Å correspond to the Ru–Co bond length within the catalyst structure. These findings indicate the presence of the Ru–O bond path and the absence of second-shell Ru–Ru interactions in $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$ (Supplementary Figs. 6 and 7).

We conducted a detailed examination of the k and R curves using wavelet transform analysis to obtain more precise peak information. This analysis provided detailed insights into the observed peaks. The results indicated that the distinctive Ru–O bond region, characteristic of the Ru single-atom catalyst, was absent in the Ru nanoparticles ($\text{Ru}_{\text{NP}}/\text{CoAlO}_x$; Fig. 2f and Supplementary Table 1), underscoring the unique bonding environment of $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$. The dispersed structure within the Al-doped cobalt oxide matrix promotes favourable interactions between Ru single atoms and the surrounding environment, which enhances the catalyst performance. The CoAlO_x support and $\text{Ru}_{\text{NP}}/\text{CoAlO}_x$ catalyst were also characterized accordingly.

Data-guided catalyst selection and initial screening

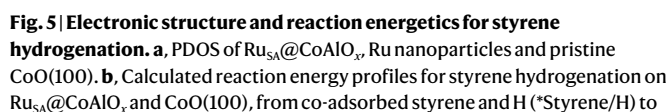
We first performed a data-driven machine learning analysis of literature-reported plastic hydrogenolysis systems to identify the key factors governing liquid fuel production. Reaction time, H_2 pressure, metal loading and catalyst identity emerged as the dominant variables influencing model-predicted yield (Fig. 3a and Supplementary Tables 2 and 3 and Figs. 8–11). Among the catalyst descriptors, Ru-based systems showed generally positive contributions. However, the literature data suggest that high yields are often obtained at high metal loadings and elevated H_2 pressure.

We therefore used the vapour-phase hydrogenation of styrene, a representative polystyrene-derived intermediate, as a probe reaction

for catalyst screening at 160 °C and 0.15 MPa in a tandem micro fixed-bed reactor operating under plug-flow conditions (Péclet number = 245; Supplementary Figs. 12–15). A broad range of catalyst supports and metal species was evaluated (Supplementary Table 4), revealing strong dependence of catalytic performance on both metal identity and metal-support interactions. Among all candidates, $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$ showed complete styrene conversion with >99% ethylcyclohexane (ECH) selectivity and a turnover number (TON) of 199.6 (Fig. 3b). Replacing Ru single atoms with nanoparticles on the CoAlO_x support reduced ECH selectivity from >99% to 20.1%. Similarly, Ru nanoparticles supported on HZSM-5 and Co_3O_4 gave comparable conversion levels but only 4.6% and 74.1% ECH selectivity, respectively. The Co/Al ratio of the support further influenced product distribution, with an optimal value of 2:1, whereas increasing Ru loading from 0.1 to 1 wt% induced C–C bond cleavage and yielded 2.7% methylcyclohexane. Despite these variations, $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$ consistently favoured benzene-ring hydrogenation, maintaining 53.7% ECH selectivity even under atmospheric pressure.

Continuous vapour-phase hydrogenation and kinetic evaluation

To further validate the catalyst performance, we scaled up the process by conducting vapour-phase hydrogenation of ethylbenzene in a continuous fixed-bed reactor using gram-scale amounts of the Ru single-atom catalyst (Supplementary Fig. 16). Ethylbenzene was chosen as the model compound to avoid kinetic inaccuracies caused by the preferential reduction of the vinyl group in styrene (Supplementary Fig. 17). Parameter studies targeting liquid hourly space velocity (LHSV), reaction pressure and temperature specified the ideal conditions for kinetic analysis (Supplementary Figs. 18 and 19). At 0.1 MPa, 1.8 h^{-1} LHSV and 160 °C, the $\text{Ru}_{\text{SA}}/\text{CoAlO}_x$ achieved steady-state within 1 h, maintaining 14.8%



We further evaluated the scalability of our approach using post-consumer PS waste, such as foam packaging, disposable cups, food containers and waste toys. The Ru_{SA}@CoAlO_x catalyst achieved over 88 wt% cycloalkanes with 92% selectivity for ECH (Fig. 6d). It also upcycled various plastics, including polyethylene (PE), polypropylene (PP), polyethylene terephthalate (PET) and their mixtures, into jet-fuel-range hydrocarbons with ideal carbon lengths (Supplementary Figs. 39–47). For instance, PE transformation yielded 85.7 wt% *n*-alkanes, predominantly C₁₀–C₁₂ (Supplementary Fig. 42). PP conversion produced 86.4 wt% *iso*-alkanes, mainly centred at C₉ (Supplementary Fig. 44). Processing mixed plastics, such as PP and PE, resulted in 85.1 wt% alkanes with 63.7 wt% in the C₈–C₁₆ jet fuel range. Other mixtures (PE/PS, PP/PS and PE/PP/PS) afforded 84.3, 82.3 and 86.6 wt% fuel-range compounds, respectively (Fig. 6e). These findings demonstrate that our strategy extends beyond PS to polyolefins such as PE and PP, whose *n*-alkane- and *iso*-alkane-rich products complement PS-derived cycloalkanes, enabling a well-balanced jet fuel composition.

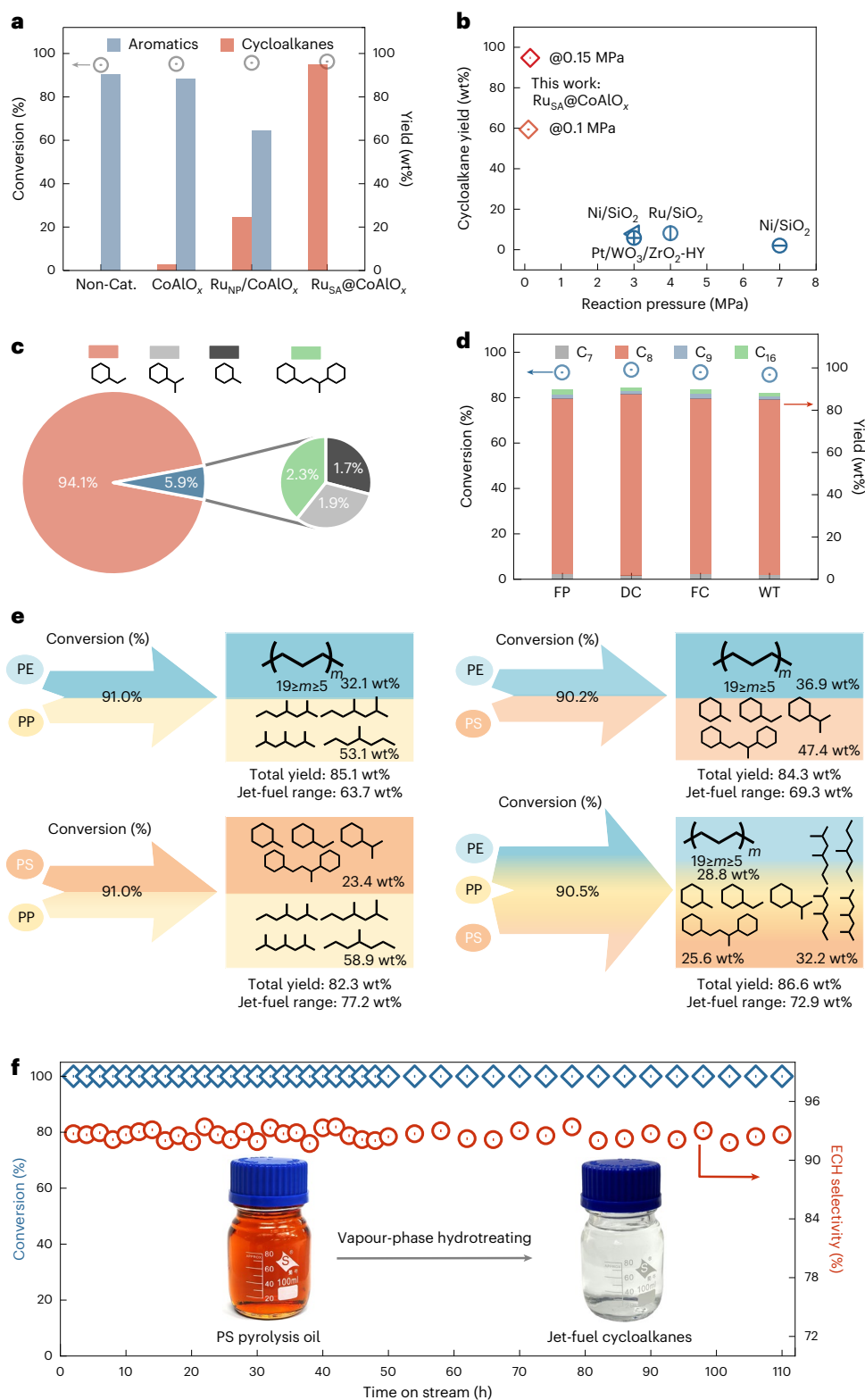


Fig. 6 | Tandem upcycling of plastic waste to jet-fuel-range hydrocarbons.

a, Effect of catalyst type on conversion and product yield during tandem upcycling of polystyrene (PS). **b**, Cycloalkane yield and product yield as a function of reaction pressure using PS as the feedstock. **c**, Product distribution (selectivity, %) of cycloalkanes obtained over Ru_{SA}@CoAlO_x. **d**, Tandem conversion of post-consumer polystyrene waste. FP, foam packaging; DC, disposable cup; FC, food container; WT, waste toy. **e**, Tandem upcycling of mixed plastic waste. **f**, Time-on-stream stability of Ru_{SA}@CoAlO_x in a bench-scale continuous fixed-bed reactor using PS pyrolysis oil. Unless

otherwise specified, experiments in **a–d** were conducted with hydropyrolysis at 460 °C and downstream hydrogenation at 160 °C under 0.15 MPa H₂, using 25 mg of catalyst and a catalyst-to-polystyrene mass ratio of 5:1. For **e**, hydropyrolysis was conducted at 500 °C and downstream vapour-phase upgrading at 210 °C under 0.15 MPa H₂, with a catalyst-to-plastic mass ratio of 5:1 and feedstock mass ratios of 2:3 for PE/PP, PE/PS and PS/PP and 1:2:2 for PE/PP/PS. For **f**, reactions were carried out at 180 °C and 0.5 MPa using 1.8 g of Ru_{SA}@CoAlO_x (bed volume, 2 ml), an LHSV of 1.8 h⁻¹ for PS pyrolysis oil and an H₂ GHSV of 1,500 h⁻¹.

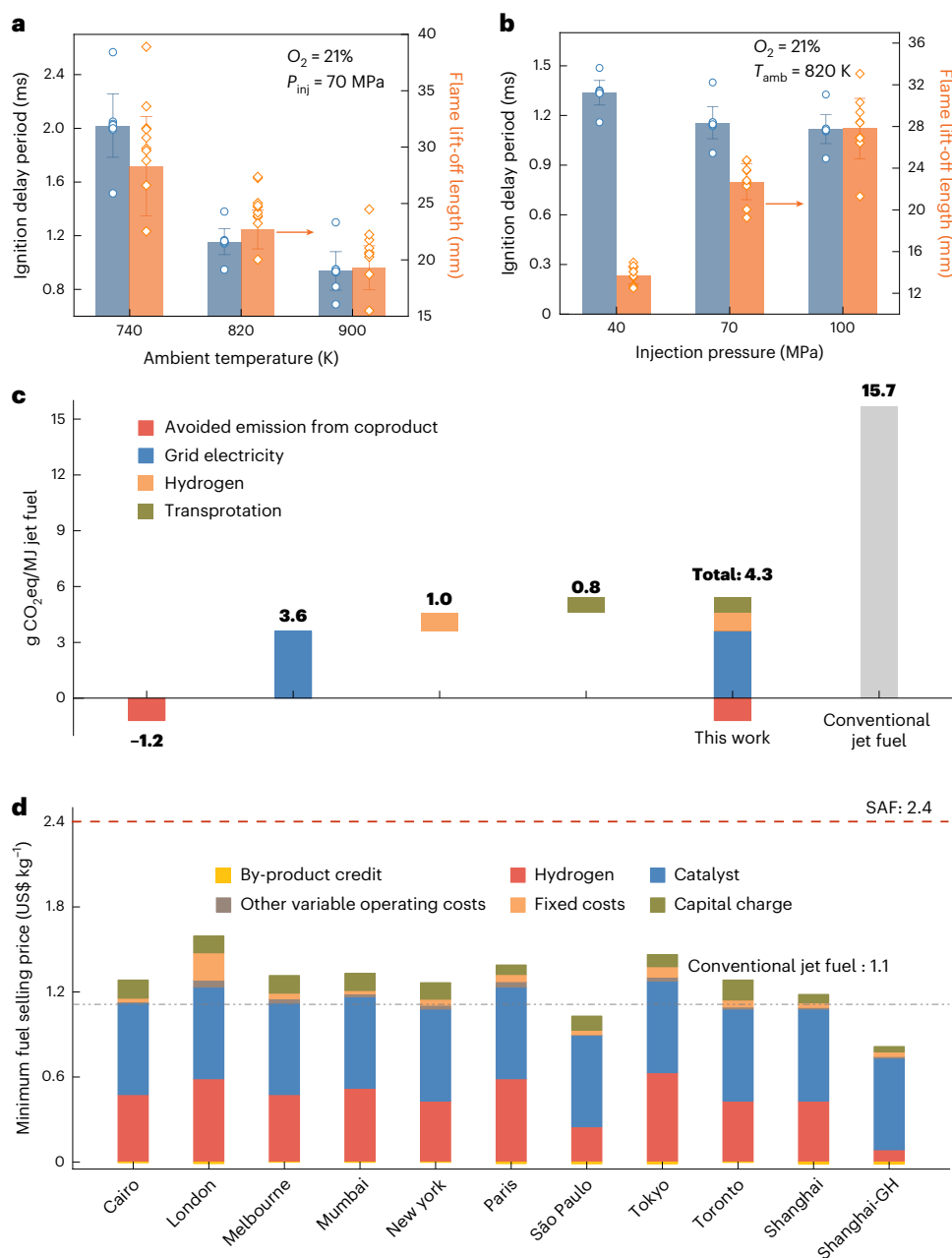


Fig. 7 | Combustion performance and sustainability assessment.

a,b, Effects of ambient temperature (**a**) and injection pressure (**b**) on ignition and combustion characteristics. **c**, Life-cycle greenhouse gas emissions of the tandem upcycling process on a per-MJ-of-jet-fuel basis. **d**, Techno-economic

analysis of the minimum fuel selling price across selected cities. Data in **a** and **b** are presented as mean $\pm$ s.d. ($n = 10$). In **d**, dashed lines indicate benchmark prices for conventional jet fuel and sustainable aviation fuel (SAF). Shanghai-GH denotes the scenario using grey hydrogen as the hydrogen source.

Stability tests for $Ru_{SA}@CoAlO_x$ were conducted using batch operations that simulated continuous feeding, where PS waste was intermittently introduced into the first-stage reactor while the catalyst in the second stage remained unchanged. This approach offers deeper insights into the long-term stability beyond previous studies focused on regeneration cycles (Supplementary Table 11)^{22,30}. Impressively, the catalyst maintained a 93.2 wt% cycloalkane yield after 160 feedings, showing excellent stability (Supplementary Figs. 48–50). Further industrial potential was assessed in a continuous fixed-bed reactor using PS pyrolysis oil under 0.5 MPa and 1.8 h^{-1} LHSV (Supplementary Fig. 51). Over 110 h of continuous operation, the catalyst maintained complete conversion of PS-derived aromatics (Fig. 6f). Heteronuclear Single Quantum Coherence spectrum confirmed the absence of aromatic peaks and the emergence of cycloalkanes in the hydrogenated

products (Supplementary Fig. 52). The feasibility for industrial scale-up was underscored by a TON of 195,800 and a space-time yield of $1,648 \text{ g/g}_{Ru}/\text{h}$ for ECH.

Combustion behaviour and sustainability

We investigated the ignition and combustion characteristics of jet fuel cycloalkane compounds, formulated specifically according to the cycloalkane product distribution derived from PS waste. Using a constant volume chamber and advanced optical diagnostics, we were able to visualize and analyse their combustion behaviour in detail (Supplementary Figs. 53–56 and Supplementary Tables 12–14). At an ambient temperature of 740 K, the tailored fuel exhibited an initial light emission approximately 1.9 ms after the start of injection, demonstrating its excellent ignition properties (Fig. 7a). When the temperature was

increased to 820 K and 900 K, the light emission occurred earlier at 1.3 ms and 0.8 ms, respectively. The higher temperatures also led to a reduced flame lift-off length, indicating stronger interaction between the flame and fuel droplets and more efficient evaporation at the injection tip, further highlighting the superior combustion efficiency. Notably, this tailored fuel ignited effectively at a much lower pressure of 40 MPa, compared to the 100 MPa or more typically required for conventional diesel (Fig. 7b).

A life-cycle assessment was conducted to evaluate well-to-pump and well-to-wheel emissions of the tandem upcycling process (Fig. 7c, Supplementary Figs. 57–59 and Supplementary Tables 15–23). The system boundary includes plastic waste collection, conversion and jet fuel distribution, while excluding the initial plastic manufacturing phase, as these emissions were already accounted for in its previous life cycle. Aspen Plus-derived mass and energy balances showed substantial chemical-energy retention across the process, with 76.5% of the total product energy concentrated in the jet fuel fraction. Producing 1 MJ of jet fuel resulted in 4.3 g of CO₂-equivalent (CO₂eq) emissions during the well-to-pump phase, which includes resource extraction, fuel production and distribution. This marks a 73% reduction in emissions compared to fossil-derived jet fuels (15.7 g CO₂eq). When considering the full life cycle, including fuel combustion in engines, well-to-wheel total greenhouse gas emissions were estimated at 77.3 g CO₂eq (Supplementary Table 20), lower than the 88.7 g CO₂eq associated with fossil-based fuels⁴⁵. Sensitivity analysis (Supplementary Fig. 60) further shows that hydrogen emissions intensity and process energy efficiency exert the strongest influence on process carbon emissions. Together, these results identify tandem upcycling as a promising route to lower carbon emissions relative to conventional fuel production, with potential relevance for decarbonizing the aviation sector.

A techno-economic assessment of low-carbon jet fuel production was performed across ten cities worldwide, determining the minimum selling price (MSP) using the discounted cash flow method (Fig. 7d and Supplementary Tables 24–38). The MSPs are comparable to conventional jet fuel, which remain well below the benchmark price for sustainable aviation fuel. Among the selected cities, São Paulo exhibited the lowest MSP at US\$1.0 kg⁻¹, benefiting from affordable green hydrogen and low electricity prices. In contrast, London and Tokyo had markedly higher MSPs (US\$1.6–US\$1.8 kg⁻¹), primarily due to elevated energy and hydrogen costs. The transformation of PS into jet-fuel-range cycloalkanes requires a substantial input of hydrogen. For example, fully hydrogenating 1 mole of styrene monomer to ECH demands 4 moles of H₂. The strong dependence on hydrogen leads to notable cost variations in Shanghai, where green hydrogen is priced at US\$1.2 kg⁻¹ and grey hydrogen at US\$0.8 kg⁻¹. Sensitivity analyses indicate that MSPs are currently most affected by fluctuations in the costs of catalysts, hydrogen and waste plastic acquisition (Supplementary Fig. 61). With advances in technology driving down the costs of catalysts and hydrogen, changes in government regulations regarding waste plastic disposal fees may shift feedstock costs to become a dominant factor in MSP determination.

Conclusion

This study demonstrates a highly efficient strategy for ambient-pressure upcycling plastic waste into jet-fuel-range cycloalkanes, addressing key challenges in aviation fuel production. By integrating hydrolysis with vapour-phase hydrogenation, our approach enables rapid and selective hydrogenation of aromatics using a Ru_{SA}@CoAlO_x catalyst. The catalyst exhibits exceptional activity, achieving a turnover frequency of 144 s⁻¹ and maintaining stability over 110 hours of continuous operation. Life-cycle and techno-economic assessments indicate that this method reduces CO₂ emissions by 73% and achieves a competitive fuel production cost as low as US\$1.0 kg⁻¹. Beyond polystyrene, this strategy is applicable to mixed polyolefins, expanding its potential for large-scale waste valorization. These findings provide a scalable

and energy-efficient route for converting plastic waste into alternative aviation fuels, offering a viable pathway towards carbon-neutral aviation. Future work will focus on further optimizing catalyst design and expanding feedstock compatibility to enhance the economic and environmental benefits of this approach.

Methods

Materials and chemicals

Polystyrene (PS, weight-average molecular weight, Mw = 280 kDa), polyethylene (Mw = 3,000 kDa) and polypropylene (Mw = 12 kDa) were sourced from Sigma-Aldrich. Post-consumer plastic waste, including foam packaging, disposable cups, food containers and toys, was collected from a supermarket and recycling centre in Nanjing, China. The plastic feedstock was characterized by elemental and proximate analyses and Fourier transform infrared spectroscopy. Analytic styrene, ethylbenzene and external calibration standards (benzene, toluene, ethylbenzene, methylstyrene, cumene, cyclohexane, methylcyclohexane and ethylcyclohexane) were obtained from Sigma-Aldrich. Additional standards, including isopropylcyclohexane, 1,3-dicyclohexylbutane, C₅–C₁₀ and C₇–C₄₀ *n*-alkanes (>99.9% purity) and *iso*-alkanes (for example, 2,4-dimethylheptane), were purchased from Mackin. Metal oxides (CeO₂, TiO₂, Co₃O₄, CoO and SiO₂) were provided by Mackin, while commercial HZSM-5 and USY zeolites were sourced from Nankai Catalyst. Ru-based catalysts (5 wt% Ru/Al₂O₃ and 5 wt% Ru/C) were purchased from Sigma-Aldrich.

Cobalt nitrate hydrate and aluminium nitrate hydrate (analytical grade) were supplied by China National Medicines Corporation Ltd. Sodium hydroxide was obtained from Beijing Chemical Reagents. Ruthenium chloride (anhydrous and trihydrate) was purchased from Adamas, and hydrochloric acid from Beijing Tongguang Fine Chemical Company. All reagents and absolute ethanol were used as received without further purification. Distilled water was degassed by boiling for 1 min.

Catalyst preparation

To prepare the Ru_{SA}@CoAlO_x catalyst, cobalt nitrate (4.2 g, 14.2 mmol) and aluminium nitrate (2.7 g, 7.1 mmol) were dissolved in 20 ml of distilled water using ultrasonic agitation for 10 min (solution A). Separately, RuCl₃·3H₂O (50 μL, 20 mg ml⁻¹) and NaOH (1.4 g) were dissolved in 19.9 ml of distilled water to form solution B. The RuCl₃·3H₂O solution was prepared by ultrasonically dissolving 20 mg of RuCl₃·3H₂O in 1 ml of distilled water. Solutions A and B were added dropwise into 40 ml of deionized water under vigorous stirring, maintaining pH at 8.5 ± 0.2 by adjusting the addition rate. After complete addition of solution A, stirring continued at room temperature for 16 h. The precipitate was washed twice with water and ethanol via centrifugation, dried and ground into a homogeneous powder. The resulting Ru_{SA}@CoAlO_x powder was subjected to 400 °C for 2 h in a 5% H₂/Ar atmosphere at a heating rate of 5 °C min⁻¹. Throughout the process, the ambient temperature was maintained at 24 °C. For scale-up vapour-phase hydrogenation experiments (1.8 g catalyst per trial), the synthesis was scaled up 20-fold, with filtration replacing centrifugation during washing.

Catalyst characterization

Crystalline structures were analysed via X-ray powder diffraction using a Rigaku RU-200b diffractometer with Cu Kα radiation (λ = 1.5406 Å). Surface chemical states were examined through X-ray photoelectron spectroscopy on an ULVAC PHI Quantera microscope. Catalyst morphology was characterized using transmission electron microscopy (TEM) with a Hitachi H-800, while high-resolution TEM, energy-dispersive spectroscopy line scans and elemental mapping were conducted on a JEOL JEM-2100F at 200 kV. High-angle annular dark-field scanning transmission electron microscopy imaging was performed on a JEOL 200 F microscope equipped with a probe aberration corrector at 200 keV. The Ru content was quantified via inductively

coupled plasma optical emission spectroscopy using an Optima 7300 DV system.

Ru k-edge X-ray absorption fine structure spectra were collected at the 1W1B beamline of the Beijing Synchrotron Radiation Facility under 2.5 GeV and a 250 mA maximum current. Measurements were conducted in fluorescence-detected mode using a 19-element Ge solid-state detector array with Mo and Fe filters. For X-ray absorption fine structure analysis, 40 mg of catalyst was ground and pressed into 8 mm pellets without binders or diluents. Reacted catalysts mixed with quartz were prepared similarly, with 60 mg ground for 10 minutes before pressing. Reference spectra for Ru foil, RuO₂ and RuCl₃ were collected in transmission mode at the 1W1B and BL14W1 beamlines.

Micro tandem fixed-bed reactor for catalytic testing

A tandem two-stage flow-through pressurized fixed-bed reactor (RX3050TR, Frontier Lab) was specifically designed for plastic waste upcycling. The system comprises two identical deactivated stainless-steel microreactors, with the first measuring 76 × 125 × 75 mm and the second 76 × 125 × 160 mm, both having an inner diameter of 4.6 mm. The reactors are vertically connected, each featuring independent temperature control, with maximum temperatures of 1,000 °C for the first-stage reactor and 600 °C for the second-stage reactor.

In the first reactor, a high-pressure solid sampler introduced plastic powders for hydrolysis, with a maximum capacity of approximately 50 mg. The second reactor featured a quartz tube (inner diameter: 3 mm, length: 78 mm) loaded with catalysts, diluted with inert silicon dioxide, for hydrogenation. Throughout the experiments, the hydrogen flow rate was consistently maintained at 80 ml min⁻¹. Catalyst loading was controlled at 25 ± 0.05 mg, maintaining a catalyst-to-feedstock mass ratio of 5:1. Before each experiment, catalysts were pre-reduced at 450 °C for 2 hours under a hydrogen atmosphere in the second reactor. A high-pressure flow controller regulated the reaction pressure (ranging from 0.1 to 3 MPa) and controlled the flow rates of both carrier and reactant gases, while a safety controller ensured safe operation under high pressures.

In the tandem reactor set-up, a styrene hydrogenation probe reaction was initially conducted, where the first-stage reactor vaporized styrene at 160 °C and the second-stage reactor, loaded with a specified amount of catalyst, facilitated catalytic hydrogenation at the same temperature under 0.15 MPa. To evaluate catalyst stability, 25 mg of Ru_{SA}@CoAlO_x was used in the second-stage reactor. Simulating a continuous feeding process, real-world PS foam was introduced into the first-stage reactor while keeping the catalyst unchanged and subjecting it to 160 feed cycles. After the reaction, the spent catalysts were collected for further characterization.

The reaction products were identified using an online gas chromatography-mass spectrometry/thermal conductivity detector system (Agilent 8890 GC-5977B MS) connected to the fixed-bed reactor, positioned outside the second reactor. Gas chromatography analysis was performed using an HP-PONA column (50 m length, 0.20 mm inner diameter, 0.50-μm film thickness). The oven was initially held at 45 °C for 1 minute, then heated to 320 °C at a rate of 10 °C min⁻¹. Mass spectrometry detection scanned *m/z* values from 2 to 550 to capture molecular ions. Product identification was performed using the National Institute of Standards and Technology mass spectral library, with external standards for calibration.

Scale-up and continuous vapour-phase hydrogenation evaluation

Benzene-ring hydrogenation was performed in a continuous two-stage fixed-bed reactor comprising a vaporizer, a hydrogenation reactor, a liquid-feed system and a condenser. The vaporizer and hydrogenation reactor were independently temperature-controlled over ranges from room temperature to 800 °C and from room temperature to 1,200 °C, respectively, to ensure complete vaporization of liquid feedstocks

(styrene or ethylbenzene) before entry into the reactor. The reaction tube was made of 316 l stainless steel (8 mm inner diameter, 16 mm outer diameter and 325 mm length) and was rated for pressures up to 10 MPa. Liquid feed was delivered by a high-pressure pump at 0.001–4.999 ml min⁻¹, with a maximum operating pressure of 42 MPa and a flow accuracy of ±0.2%.

The performance of the Ru_{SA}@CoAlO_x catalyst was evaluated using ethylbenzene as the feedstock. Typical reaction conditions were 160 °C and atmospheric pressure, with 1.8 g of catalyst mixed with quartz sand to a total bed volume of 2 ml. Hydrogen was supplied at 50 ml min⁻¹, corresponding to a gas hourly space velocity (GHSV) of 1,500 h⁻¹. A 10 wt% ethylbenzene solution in hexane was fed at 36 ml min⁻¹, corresponding to a liquid hourly space velocity (LHSV) of 1.8 h⁻¹. The stability of Ru_{SA}@CoAlO_x was then assessed under continuous operation using PS-derived pyrolysis oil at 180 °C and 0.5 MPa for 110 h. The pyrolysis oil was prepared by pyrolysing PS in a fixed-bed reactor (600 mm length, 50 mm diameter) at 480 °C with a carrier-gas flow rate of 50 ml min⁻¹. Each batch processed 5 g of PS, yielding 90.7 wt% liquid oil, 5.4 wt% solid char and 3.9 wt% gas. Repeated experiments produced 400 g of pyrolysis oil for the continuous gas-phase catalytic hydrogenation test.

Operando DRIFTS experiments

Operando DRIFTS experiments were conducted using a Fourier transform infrared spectrometer (Bruker, Vertex 80 V) equipped with a mercury cadmium telluride detector to monitor the vapour-phase hydrogenation process in real time. The catalyst was first pretreated under H₂ flow (30 ml min⁻¹) at 450 °C for 2 h, followed by cooling to 160 °C. A background spectrum was recorded at 160 °C in H₂ flow. The catalyst was then exposed to a 10% styrene mixture in H₂ at 160 °C and 0.1 MPa for 50 min, during which spectra of surface species were collected. Operando DRIFT spectra were acquired with 16 scans at a resolution of 4 cm⁻¹.

Machine learning study

Machine learning (ML) was employed to assess the effects of reaction time, pressure, catalyst metal type and metal loading on liquid fuel yield during plastic waste hydrogenolysis. Data were extracted from Web of Science using the keywords 'plastic waste' and 'hydrogenolysis' to compile a dataset of relevant journal articles published between January 2020 and June 2024. The dataset included experimental conditions and reported yields, with doping metals numerically encoded via one-hot encoding (Supplementary Table 1).

Twelve ML algorithms were trained and optimized using hyperparameter tuning in Python 3.8.8 with Scikit-Learn 1.2.2. Model performance was evaluated through fivefold cross validation, and the best-performing algorithm was selected based on the mean test score. Feature attribution analysis was conducted using SHAP (v0.41.0) to quantify each parameter's contribution to liquid fuel yield predictions. Further details are provided in Supplementary Methods.

Density functional theory

On the basis of combined X-ray diffraction, X-ray photoelectron spectroscopy and X-ray absorption spectroscopy analyses, a representative structural model of Ru single atoms stabilized on Al-moderately substituted CoO surfaces (Ru_{SA}@CoAlO_x) was constructed. Density functional theory (DFT) calculations were performed using the Vienna Ab-initio Simulation Package⁴⁶, employing the projector-augmented wave method and the generalized gradient approximation with the Perdew–Burke–Ernzerhof exchange-correlation functional; van der Waals interactions were accounted for via Grimme's D3 correction. For the slab model, a three-layer CoO(100) configuration was employed, with its top two layers allowed to relax. The strong correlation effects of localized Co 3*d* electrons were addressed using the DFT + U method (*U*_{eff} = 3.7 eV). Transition states along the reaction pathway were located using the Climbing Image Nudged Elastic Band method⁴⁷.

Detailed descriptions of the computational methods and models are provided in Supplementary Methods.

Combustion performance evaluation

A constant volume combustion chamber was used to evaluate the combustion performance of jet fuel cycloalkane model compounds, formulated based on the product distribution from PS plastics. The chamber was capable of reaching a maximum temperature of 900 K and an ambient back pressure of up to 6 MPa, simulating top dead centre conditions in engines. A high-pressure fuel injection system facilitated fuel delivery, with a single-hole ejector (0.12 mm nozzle diameter) positioned at the chamber top. The spray combustion test system incorporated a Bayonet Nut Connector synchronizer for synchronized control of the injector and camera. An Intensified Charge-Coupled Device camera equipped with a 310 ± 10 nm band-pass filter measured flame lift-off length, while a high-speed camera captured ignition delay and flame evolution (Supplementary Methods). To ensure reliability, each test condition was evaluated based on an average of ten measurements.

Life-cycle assessment

To assess the environmental performance of converting plastic waste into jet fuel, mass and energy balances were first established from elemental analyses and Aspen Plus simulations (Supplementary Tables 15–17). These balances provided data on feedstock conversion, utility requirements and intermediate stream compositions, which were used as inputs for the subsequent life-cycle greenhouse gas emissions analysis. A life-cycle assessment was then conducted under well-to-pump and well-to-wheel frameworks, encompassing waste transport, refining, product distribution and fuel combustion (Supplementary Tables 18–23 and Supplementary Figs. 58 and 59). Because the feedstock comprises post-consumer plastics, upstream manufacturing emissions were excluded to represent an end-of-life scenario. Global warming impacts were evaluated over a 100-year time horizon using two functional units: treatment of one tonne of waste plastic and delivery of one megajoule (MJ) of jet fuel. Sensitivity analyses considered variations in electricity intensity, hydrogen source and transport distance to capture potential emission changes under different operational and regional conditions. Full datasets, assumptions and emission calculations are provided in the Supplementary Methods.

Techno-economic analysis

A techno-economic analysis was conducted to evaluate the economic feasibility of converting plastic waste into jet fuel. For each scenario, the minimum selling price (MSP) required to achieve the target internal rate of return was determined using a discounted cash flow model that accounted for capital recovery, debt service, depreciation and taxation. Capital expenditures were estimated from scaled equipment costs using established cost indices and further included engineering, contingency and start-up costs. Operating expenditures included feedstock, catalyst replacement, utilities, labour and maintenance. Ten major cities were selected to capture variation in waste generation, energy prices, labour costs and landfill-diversion policies.

Two hydrogen supply pathways, grey and green, were considered to assess the economic impact of low-carbon hydrogen. Sensitivity analysis was performed by independently varying key parameters, including jet fuel yield, feedstock cost, hydrogen price, process energy efficiency, tax policy, equity financing ratio and catalyst consumption rate, to quantify their effects on MSP. Detailed assumptions, scenario-specific results and calculation procedures are provided in the Supplementary Methods.

Data evaluation

Plastic waste conversion was calculated according to equation (1):

$$\text{Conversion} = \frac{m_0 - m_s}{m_0} \times 100\% \quad (1)$$

where m_0 is the initial mass of the plastic waste and m_s is the mass of the solid residue after reaction.

The yield of liquid hydrocarbons was calculated using equation (2):

$$y_i = \frac{m_i}{m_0} \times 100\% \quad (2)$$

where m_i represents the mass of the specific product.

The selectivity of hydrocarbons was calculated as follows:

$$S_i = \frac{m_i}{\sum m_j} \times 100\% \quad (3)$$

where $\sum m_j$ represents the total mass of liquid hydrocarbons.

The turnover frequency (TOF) of ethylcyclohexane (ECH) over Ru-based catalysts was determined using the following equation:

$$\text{TOF}_{\text{ECH}} = \frac{\text{mol of ECH}}{\text{mol of Ru} \times \tau} \quad (4)$$

where τ represents the residence time, defined as the inverse of the gas hourly space velocity of ethylbenzene vapour at 160 °C and 0.1 MPa. The total feedstock, a 10% ethylbenzene solution, was 144 ml with a feed rate of 36 ml h^{−1}.

The turnover number (TON) over the 110-hour Ru_{SA}@CoAlO_x stability test was calculated as follows:

$$\text{TON} = \frac{M_{\text{PS-oil}} \times \text{Conversion}}{M_{\text{Ru}}} \quad (5)$$

where $M_{\text{PS-oil}}$ is the total mass of PS pyrolysis oil, M_{Ru} is the mass of Ru metal in the Ru_{SA}@CoAlO_x catalyst.

The space time yield (STY) of ECH was calculated as:

$$\text{STY}_{\text{ECH}} = \frac{M_{\text{PS-oil}} \times \text{Conversion} \times \text{ECH}_{\text{selectivity}}}{M_{\text{Ru}} \times \text{reaction time}} \quad (6)$$

To evaluate whether the tandem fixed-bed reactor achieves plug-flow conditions, enabling accurate residence time calculation, we analysed the Péclet number (P_e) along with convective and diffusive timescales. A reactor satisfying the P_e and relevant dimensionless group criteria can be reasonably assumed to exhibit plug flow, characterized by negligible axial mixing and complete transverse mixing. The P_e number is defined as:

$$P_e = R_e \times S_c = \frac{\rho u D}{\mu} \times \frac{\mu}{\rho D_{\text{AB}}} = \frac{u D}{D_{\text{AB}}} \quad (7)$$

The characteristic diffusion timescale ($\tau_{\text{sp-diff},R}$) is defined as:

$$\tau_{\text{sp-diff},R} = \frac{R^2}{D} \quad (8)$$

The characteristic convection timescale ($\tau_{\text{conv},L}$) is defined as:

$$\tau_{\text{conv},L} = \frac{L}{\bar{v}} \quad (9)$$

The activation energy (E_a) for the hydrogenation of ethylbenzene to ECH was determined using the Arrhenius equation:

$$k = A \exp\left(-\frac{E_a}{RT}\right) \quad (10)$$

where k is the reaction rate constant, and A is the pre-exponential (or frequency) factor. E_a is the activation energy, R is the ideal gas constant (8.314 J (mol·K)^{−1}), and T is the absolute temperature in Kelvin (K).

For the combustion behaviour of the plastic-derived model fuel, the physical and chemical parameters of the mixed plastic-derived fuel were estimated using the following formula:

$$\phi = \sum_i^n x_i \phi_i \quad (11)$$

where ϕ represents the physical and chemical parameters of the mixed fuel, x_i is the mass fraction of each fuel component, and ϕ_i denotes the physical and chemical parameters of each component.

The ignition delay period is determined using criteria based on light intensity measurements, as specified in equations (12)–(14):

$$I_{\text{back}} = \sum_i^{10} \frac{I_i}{i} \quad (12)$$

where I_{back} represents the average luminous intensity of the initial ten images following the start of fuel injection, with each I denoting the luminous intensity of an individual image, and i indicating the image index.

$$I_i^{\text{peak}} \geq 2.25 \times I_{\text{back}} \quad (13)$$

The maximum light intensity of an image, denoted as I_i^{peak} , is identified by its corresponding index i when it meets the criteria specified in equation (11).

$$\text{ID} = \frac{1}{\nu} \times i - \text{SOI} \quad (14)$$

where ν represents the frame rate, and i is determined according to equation (11).

Data availability

All data supporting the findings of this study are available within the Article, its Supplementary Information and the accompanying Source Data files. Source data are provided with this paper.

References

- Li, G. et al. Production of Renewable Hydrocarbon Biofuels with Lignocellulose and Its Derivatives over Heterogeneous Catalysts. *Chem. Rev.* **124**, 2889–2954 (2024).
- Sun, D. et al. Theoretic heat sink simulation and experimental investigation of the pyrolysis of substituted cyclohexanes. *J. Energy Chem.* **24**, 119–125 (2015).
- Li, Y., Jin, B., Zhang, X. & Liu, G. Pyrolysis and heat sink of an endothermic hydrocarbon fuel EHf-851. *J. Anal. Appl. Pyrol.* **155**, 105084 (2021).
- Jia, T. et al. A comprehensive review of the thermal oxidation stability of jet fuels. *Chem. Eng. Sci.* **229**, 116157 (2021).
- Hwang, K. C. & Sagadevan, A. One-pot room-temperature conversion of cyclohexane to adipic acid by ozone and UV light. *Science* **346**, 1495–1498 (2014).
- Datta, S. et al. Self-assembled poly-catenanes from supramolecular toroidal building blocks. *Nature* **583**, 400–405 (2020).
- Chen, L. et al. Reversible dehydrogenation and rehydrogenation of cyclohexane and methylcyclohexane by single-site platinum catalyst. *Nat. Commun.* **13**, 1092 (2022).
- Allendorf, M. D. et al. Challenges to developing materials for the transport and storage of hydrogen. *Nat. Chem.* **14**, 1214–1223 (2022).
- Mehdad, A., Jentoft, R. E. & Jentoft, F. C. Single-phase mixed molybdenum-niobium carbides: synthesis, characterization and multifunctional catalytic behavior in toluene conversion. *J. Catalysis* **351**, 161–173 (2017).
- Liu, S., Kots, P. A., Vance, B. C., Danielson, A. & Vlachos, D. G. Plastic waste to fuels by hydrocracking at mild conditions. *Sci. Adv.* **7**, eabf8283 (2021).
- Lee, K., Jing, Y., Wang, Y. & Yan, N. A unified view on catalytic conversion of biomass and waste plastics. *Nat. Rev. Chem.* **6**, 635–652 (2022).
- Singh, A. et al. Techno-economic, life-cycle, and socioeconomic impact analysis of enzymatic recycling of poly(ethylene terephthalate). *Joule* **5**, 2479–2503 (2021).
- Jehanno, C. et al. Critical advances and future opportunities in upcycling commodity polymers. *Nature* **603**, 803–814 (2022).
- Haussler, M., Eck, M., Rothauer, D. & Mecking, S. Closed-loop recycling of polyethylene-like materials. *Nature* **590**, 423–427 (2021).
- Vollmer, I., Jenks, M. J. F., Mayorga Gonzalez, R., Meirer, F. & Weckhuysen, B. M. Plastic waste conversion over a refinery waste catalyst. *Angew. Chem. Int. Ed.* **60**, 16101–16108 (2021).
- European Parliament & Council of the European Union. Regulation (EU) 2023/2405 of the European Parliament and of the Council of 18 October 2023 on ensuring a level playing field for sustainable air transport (ReFuelEU Aviation). *Official Journal of the European Union L*, 2023/2405 <https://eur-lex.europa.eu/eli/reg/2023/2405/oj/eng> (2023).
- Sun, J. A. et al. Size and structure effects of carbon-supported ruthenium nanoparticles on waste polypropylene hydrogenolysis activity, selectivity, and product microstructure. *ACS Catal.* **14**, 3228–3240 (2024).
- Celik, G. et al. Upcycling single-use polyethylene into high-quality liquid products. *ACS Cent. Sci.* **5**, 1795–1803 (2019).
- Kots, P. A. et al. Electronic modulation of metal-support interactions improves polypropylene hydrogenolysis over ruthenium catalysts. *Nat. Commun.* **13**, 5186 (2022).
- Tan, T. et al. Upcycling plastic wastes into value-added products by heterogeneous catalysis. *ChemSusChem* **15**, e202200522 (2022).
- Nakaji, Y. et al. Low-temperature catalytic upgrading of waste polyolefinic plastics into liquid fuels and waxes. *Appl. Catal. B* **285**, 119805 (2021).
- Zhang, F. et al. Polyethylene upcycling to long-chain alkylaromatics by tandem hydrogenolysis/aromatization. *Science* **370**, 437–441 (2020).
- Rorrer, J. E., Beckham, G. T. & Roman-Leshkov, Y. Conversion of polyolefin waste to liquid alkanes with Ru-based catalysts under mild conditions. *JACS Au* **1**, 8–12 (2021).
- Li, R. & Wang, D. Understanding the structure-performance relationship of active sites at atomic scale. *Nano Res.* **15**, 6888–6923 (2022).
- Gan, T. & Wang, D. Atomically dispersed materials: ideal catalysts in atomic era. *Nano Res.* **17**, 18–38 (2024).
- Zheng, X., Li, B., Wang, Q., Wang, D. & Li, Y. Emerging low-nuclearity supported metal catalysts with atomic level precision for efficient heterogeneous catalysis. *Nano Res.* **15**, 7806–7839 (2022).
- Li, L. et al. Converting plastic wastes to naphtha for closing the plastic loop. *J. Am. Chem. Soc.* **145**, 1847–1854 (2023).
- Wang, N. M. et al. Chemical recycling of polyethylene by tandem catalytic conversion to propylene. *J. Am. Chem. Soc.* **144**, 18526–18531 (2022).
- Tennakoon, A. et al. Catalytic upcycling of high-density polyethylene via a processive mechanism. *Nat. Catal.* **3**, 893–901 (2020).
- Du, J. et al. Efficient solvent- and hydrogen-free upcycling of high-density polyethylene into separable cyclic hydrocarbons. *Nat. Nanotechnol.* **18**, 772–779 (2023).

31. Duan, J. et al. Coking-resistant polyethylene upcycling modulated by zeolite micropore diffusion. *J. Am. Chem. Soc.* **144**, 14269–14277 (2022).
32. Ellis, L. D. et al. Tandem heterogeneous catalysis for polyethylene depolymerization via an olefin-intermediate process. *ACS Sustain. Chem. Eng.* **9**, 623–628 (2021).
33. Jie, X. et al. Microwave-initiated catalytic deconstruction of plastic waste into hydrogen and high-value carbons. *Nat. Catal.* **3**, 902–912 (2020).
34. Conk, R. J. et al. Catalytic deconstruction of waste polyethylene with ethylene to form propylene. *Science* **377**, 1561–1566 (2022).
35. Sullivan, K. P. et al. Mixed plastics waste valorization through tandem chemical oxidation and biological funneling. *Science* **378**, 207–211 (2022).
36. Dong, Q. et al. Depolymerization of plastics by means of electrified spatiotemporal heating. *Nature* **616**, 488–494 (2023).
37. Yang, R. X., Jan, K., Chen, C. T., Chen, W. T. & Wu, K. C. Thermochemical conversion of plastic waste into fuels, chemicals, and value-added materials: a critical review and outlooks. *ChemSusChem* **15**, e202200171 (2022).
38. Wang, Z., Burra, K. G., Lei, T. & Gupta, A. K. Co-pyrolysis of waste plastic and solid biomass for synergistic production of biofuels and chemicals—a review. *Progr. Energy Combust. Sci.* **84**, 100899 (2021).
39. Wan Mahari, W. A. et al. Generating alternative fuel and bioplastics from medical plastic waste and waste frying oil using microwave co-pyrolysis combined with microbial fermentation. *Renew. Sust. Energ. Rev.* **153**, 111790 (2022).
40. Miao, Y. et al. Photothermal recycling of waste polyolefin plastics into liquid fuels with high selectivity under solvent-free conditions. *Nat. Commun.* **14**, 4242 (2023).
41. Gao, J. et al. Ambient hydrogenation and deuteration of alkenes using a nanostructured Ni-core-shell catalyst. *Angew. Chem. Int. Ed.* **60**, 18591–18598 (2021).
42. Zeng, L. et al. Recycling valuable alkylbenzenes from polystyrene through methanol-assisted depolymerization. *Angew. Chem. Int. Ed.* **63**, e202404952 (2024).
43. Vance, B. C., Najmi, S. & Vlachos, D. G. Polystyrene hydrogenolysis to high-quality lubricants using Ni/SiO₂. *ACS Catal.* **14**, 5389–5402 (2024).
44. Vance, B. C., Kots, P. A., Wang, C., Granite, J. E. & Vlachos, D. G. Ni/SiO₂ catalysts for polyolefin deconstruction via the divergent hydrogenolysis mechanism. *Appl. Catal. B* **322**, 122138 (2023).
45. Jing, L. et al. Understanding variability in petroleum jet fuel life cycle greenhouse gas emissions to inform aviation decarbonization. *Nat. Commun.* **13**, 7853 (2022).
46. Kresse, G. & Hafner, J. Ab initio molecular dynamics for liquid metals. *Phys. Rev. B* **47**, 558–561 (1993).
47. Momma, K. & Izumi, F. VESTA 3 for three-dimensional visualization of crystal, volumetric and morphology data. *J. Appl. Crystallogr.* **44**, 1272–1276 (2011).
48. Wang, J. et al. Breaking the yield-selectivity trade-off in polystyrene waste valorization via tandem depolymerization and hydrogenolysis. *Nat. Nanotechnol.* **21**, 87–94 (2026).

Acknowledgements

This work was supported by the National Natural Science Foundation of China (22325101, D.W.; 52376195, J.W.), the NSFC Center for Single-Atom Catalysis (22388102, Y.L.; J.L.) and the National Key R&D Program of China (2023YFA1506801, D.W.; 2023YFA1506600, J.L.). We acknowledge Z. He at Jiangsu University for assistance with the combustion tests and J. Liu at Dalian University of Technology for support with the in situ diffuse reflectance infrared characterization.

Author contributions

Y.L., J.J., J.L. and D.W. supervised the project. J.W., Z.Z., D.L., Y.-F.J., H. Z. and W.Z. developed the methodology. J.W., Z.Z., Y.Z., S.L. and S.W. conducted the investigation. J.L., Y.-F.J., H.Z., W.Z. and S.W. performed the theoretical calculations. Y.Z. and Z.Z. carried out the morphology characterization. J.W., S.W., W.Z., Y.-F.J. and Z.Z. drafted the manuscript. D.L.S., D.W., J.L., J.J. and Y.L. reviewed and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-026-02078-7>.

Correspondence and requests for materials should be addressed to Zedong Zhang, Dingsheng Wang, Jun Li, Jianchun Jiang or Yadong Li.

Peer review information *Nature Energy* thanks Michiel Dusselier, Marianthi Ierapetritou, Gartzen López and the other anonymous reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

¹Jiangsu Co-Innovation Center for Efficient Processing and Utilization of Forest Resources, Nanjing Forestry University, Nanjing, China. ²Department of Chemistry, Tsinghua University, Beijing, China. ³Department of Environmental Science, Policy and Management, University of California, Berkeley, CA, USA. ⁴BioEcoAgro Joint Research Unit, INRAE, Université de Liège, Université de Lille, Université de Picardie Jules Verne, Barenton-Bugny, France. ⁵Fundamental Science Center of Rare Earths, Ganjiang Innovation Academy, Chinese Academy of Sciences, Ganzhou, China. ⁶Department of Chemistry and Guangdong Provincial Key Laboratory of Catalytic Chemistry, Southern University of Science and Technology, Shenzhen, China. ⁷School of Chemistry and Chemical Engineering, Zhengzhou Normal University, Zhengzhou, China. ⁸Institute of intelligent innovation, Henan Academy of Sciences, Zhengzhou, China. ⁹Institute of Chemical Industry of Forest Products, Chinese Academy of Forestry, Nanjing, China. ¹⁰School of Energy and Power Engineering, Jiangsu University, Zhenjiang, China. ¹¹School of Chemistry and Chemical Engineering, Qufu Normal University, Qufu, China. ¹²Department of Chemistry and Engineering Research Center of Advanced Rare-Earth Materials of Ministry of Education, Tsinghua University, Beijing, China. ¹³These authors contributed equally: Jia Wang, Zedong Zhang, Shule Wang, Ya-Fei Jiang, He Zhou, Wenhui Zhong. ✉ e-mail: zedong.zhang@northwestern.edu; wangdingsheng@mail.tsinghua.edu.cn; junli@tsinghua.edu.cn; jiangjc@icfip.cn; ydli@mail.tsinghua.edu.cn